# Donghyun Kim

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## Research Interests

**3D Vision:** Point Cloud Processing, 3D Reconstruction, Point Cloud Restoration.

**Human-Centric Vision:** Human Performance Capture, Novel-View Human Animation, 3D Human Pose Analysis, Video-Based Inverse Dynamics.

## Education

**Yonsei University** — **Seoul, South Korea**
*Ph.D. in Computer Science (Advisor: Prof. Seong Jae Hwang)* — Sep 2023 – Present

**Yonsei University** — **Seoul, South Korea**
*B.S. in Computer Science* — Mar 2019 – Aug 2023

## Publications

**Point Clouds in the Frequency Domain: Color in Amplitude, Geometry in Phase**
Donghyun Kim, Chanyoung Kim, Junhyeok Kim, Eunseo Jeong, Hyunah Ko, Youngjoong Kwon, Seong Jae Hwang
*Under Review*, 2026.

**Delaunay Canopy: Building Wireframe Reconstruction from Airborne LiDAR Point Clouds via Delaunay Graph**
Donghyun Kim, Chanyoung Kim, Youngjoong Kwon, Seong Jae Hwang
**ECCV 2026** [Paper] | [Project]

**How Noisy Poses Break Inverse Dynamics: Analysis and Mitigation for Video-Based Joint Torque Estimation**
Donghyun Kim, Chanyoung Kim, Eunseo Jeong, Youngjoong Kwon, Seong Jae Hwang
**CVPRW 2026** (PhysHuman) [Paper]

**Rethinking Graph Convolution for 2D-to-3D Hand Pose Lifting**
Chanyoung Kim, Donghyun Kim, Dong-Hyun Sim, Seong Jae Hwang, Youngjoong Kwon
**CVPRW 2026** (PhysHuman) **Oral Presentation** [Paper]

**FALCON: Fast Image Haze Removal Leveraging Continuous Density Mask**
Donghyun Kim, Seil Kang, Seong Jae Hwang
**CVPRW 2025** (NTIRE) [Paper]

**PLATYPUS: Progressive Local Surface Estimator for Arbitrary-Scale Point Cloud Upsampling**
Donghyun Kim, Hyeonkyeong Kwon, Yumin Kim, Seong Jae Hwang
**AAAI 2025** **Oral Presentation (Top 4.6%)** [Paper] | [Project]

**WoLF: Wide-scope Large Language Model Framework for CXR Understanding**
Seil Kang, Donghyun Kim, Junhyeok Kim, Seong Jae Hwang
*arXiv preprint*, 2024. [Paper]

## Research & Industry Experience

**Emory University (Emory Vision and Graphics Lab)** — **Atlanta, GA, United States**
*External Collaborator* — Jul 2026 – Present
*Visiting Researcher (Advisor: Prof. Youngjoong Kwon)* — Feb 2026 – Jun 2026
– Ongoing projects on Human Performance Capture and Novel-View Human Animation

**Arizona State University (ASU)** **Tempe, AZ, United States**
*Short-Term Scholar* Feb 2024

**Yonsei University (Medical Imaging and Computer Vision Lab)** **Seoul, South Korea**
*Ph.D. Student (Advisor: Prof. Seong Jae Hwang)* Sep 2023 – Present
*Undergraduate Student Researcher* Jul 2022 – Aug 2023
– Point Cloud Processing, 3D Reconstruction, Point Cloud Restoration, Image Dehazing, CXR Understanding

**Innoyard** **Seoul, South Korea**
*Researcher & Software Engineer* Jun 2021 – Aug 2022
– Project "InnoScan": Mobile 3D human body data capturing service using 3D point cloud registration and surface reconstruction

**Yonsei University (Dependable Computing Lab)** **Seoul, South Korea**
*Undergraduate Student Researcher (Advisor: Prof. Kyoungwoo Lee)* Jan 2021 – Jun 2021
– Project "FluidTrack" (with Severance Children's Hospital): Child-parent collaborative tracking system for pediatric voiding dysfunction

## 🏛 Scholarly Activities & Services

**Program Committee**
ECCV 2026 MoCha Workshop

**Conference Reviewer**
NeurIPS (2026 **Top Reviewer**), CVPR (2026), ECCV (2026), ICCV (2025), WACV (2024, 2025), 3DV (2026)

**Journal Reviewer**
IEEE Transactions on Image Processing (TIP), Pattern Recognition (PR)

**Teaching Assistant**
*Yonsei University (Introduction to Deep Learning and Application)* Spring 2024, Spring 2025, Spring 2026

**Invited Lecturer**
*Samsung Electronics (Introduction to Data Science)* Sep 2023 – Oct 2023

## 🏆 Honors & Awards

Graduation Project **Grand Prize** Jun 2023
*Yonsei University*

Graduation Project **Best Demo Award** Dec 2022
*Yonsei University*

Coding Challenge **2nd Prize** Dec 2021
*Korea Institute of Startup & Entrepreneurship Development (KISED)*